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Peikai Qi

Objective

I am currently a postdoc at the Morningside Center of Mathematics, Chinese Academy of Science, Beijing. My research interest is around elliptic curves, modular forms, class groups, Iwasawa modules, and Massey products.

Positions Held

2025-now **Postdoc in Mathematics**, *Morningside Center of Mathematics, Chinese Academy of Science*, Advisor: Xin Wan

Education

2019-2025 **Ph.D. in Mathematics**, *Michigan State University*, Advisor: Preston Wake

2015-2019 **Bachelor of Mathematics**, *Nankai University*

Award and Scholarship

2024 TA Award for Excellence in Teaching

2024 Dissertation Completion Fellowship

2023 Herbert T. Graham Scholarship Fund

Writings

2026 On the Quotient of a Pseudo-null Module (submitted)

2026 On Pseudonullity Conjecture for fine Selmer group (Joint with Ruichen XU, submitted)

2025 Machine Learning and LLM-Boost Symbolic Regression for Predicting $\mathbb{Q}$-Gonality of Modular Curves(Joint with Xu Zhuang, Yuxiang Yao,Po-Chu Hsu,Xiaokang Wang, accept 2025)

2024 An Analogue of Greenberg's Conjecture for CM Fields(Joint with Matt Stokes, accepted, Acta Arithmetica)

2024 p rationality and Iwasawa lambda invariant of a certain family of real quadratic fields(Joint with Matt Stokes, Submitted)

2023 Iwasawa λ invariant and Massey products(Submitted)

Papers with Undergraduate Students

- Combinatorial and Analytical Properties of Higher-Order Recursive Polynomial Sequences(2023), joint with Luke Bridges, Max Budnick, Zachary Martin, Janine Wang, and Aklilu Zeleke. Submitted.

- Multivariate Fibonacci-like Polynomials and their Applications(2022), joint with Sejin Park, Etienne Phillips, Zhan Zhan, and Ilir Ziba.
- Universally uniformly continuous metric spaces(2017), joint with Katrina Gensterblum, and Willie Wong.

Referee

- Conference Proceedings of Women in Numbers 6
- Ramanujan Journal
- Canadian Journal of Mathematics

Invited talk

Jan.22 2026	Ice city, Iwasawa invariants and Massey products
Sep.20 2025	AMSS, Iwasawa invariants, Massey products and Pseudo-null module
Nov.30 2024	CMS winter meeting, An analogue of Greenberg pseudo-null conjecture for CM fields
Nov.14 2024	Stockholm University, online, Iwasawa λ invariant and Massey products
Oct.15 2024	The University of Texas at Austin, algebra seminar, Iwasawa λ invariant and Massey products
Nov.13 2023	"Iwasawa λ invariant and Massey product", Fields institution number theory seminar, Video,`https://www.youtube.com/watch?v=82uHJV1WaOw`
Feb.20 2023	Michigan State University, Algebra seminar, Iwasawa lambda-invariants and Massey products

Conference(*presentation*)

Jun.10-Jun.14 2024	Talk for "Iwasawa λ invariant and Massey product", Canadian Number Theory Association XVI, Feilds institute, Canada
May.31-Jun.02 2024	Talk for "An analogue to Grenberg conjecture",9th Annual Graduate Student Conference in Algebra, Geometry, and Topology, Temple University, USA
May.20-May.24 2024	Talk for "Iwasawa λ invariant and Massey product",36th Automorphic Forms Workshop, Oklahoma State University, USA
May.07-May.10 2024	Poster presentation for "Iwasawa λ invariant and Massey product", Local Systems in Algebraic Geometry, Ohio State University, USA
Nov.11-Nov.13 2023	Talk for "Iwasawa λ invariant and Massey product", Binghamton University Graduate Combinatorics, Algebra, and Topology Conference, USA
Nov.04-Nov.05 2023	Poster presentation for "Iwasawa λ invariant and Massey product", Western Algebraic Geometry Symposium, Washington University in St. Louis, USA
Oct.13-Oct.15 2023	Poster presentation for "Iwasawa λ invariant and Massey product", Algebraic Geometry Northeastern Section, University of Pennsylvania, USA
Oct.7-Oct.8 2023	Poster presentation for "Iwasawa λ invariant and Massey product", MAGNTS 2023 University of Michigan,USA

Jul.31-Aug.4 2023	Lightning talk for "Iwasawa λ invariant and Massey product", Number Theory and Combinatorics in Duluth, USA
May.26-May.28 2023	Talk for "Iwasawa λ invariant and Massey product", 2023 Graduate Student Conference in Algebra, Geometry, and Topology-Philadelphia, USA
July.8 2022	Summer Undergraduate Michigan Mathematics Research (SUMMR) Confererence-Dearborn,USA

Student seminar talk(given)

Eigenvariety-Peking University

April.29 2026	Case of GL_1, classical, overconvergent automorphic forms.

Gross-Zagier formula-Peking University

Oct.10-17 2025	Weil representations and Waldspurger formula

Student number theory-MSU

Feb.9 2024	Almost Purity Theorem
Oct.4 2023	Arithmetic Fuchsian Group
Sep.13 2023	Modular Curve
Feb.09 2023	Banach space over $\mathbb{Q}_p$
Mar.16 2023	Everywhere convergent formal series and Riesz's theory II
Sep.29 2022	Hecke algebra and Atkin-Lehner-Li's theory
Sep.22 2022	Modular form and Hecke operator1
Mar.21 2022	Alpha multiplication: functorial definition
Mar.14 2022	Alpha multiplication: first approach
Oct.26 2021	The reflex field of CM-pairs
Nov.2 2021	The reflex field of CM-pairs, II
Oct.19 2020	Algebraic function fields and global function fields

Student algebra seminar-MSU

Feb.20 2024	Knots and Primes
Oct.6 2023	Introduction to Brauer Manin obstruction
Feb.17 2022	Various versions of modularity theorem
Oct.1 2021	structure of finite generated Lambda module and its adjoint

Alegebra learning seminar-MSU

Apr.05 2023	Galois categories

Student Seminar(Organized)

FS 2022-SS 2023	Student number theory in MSU - eigenvariety
FS 2023	Student number theory in MSU - Shimura Curve

Mentor

May.15-Jul.7 2023 MSU REU Program. Working as mentor assistant to help Prof Aklilu Zeleke to mentor undergraduate students Luke Bridges, Max Budnick, Zachary Martin, and Janine Wang. We worked on the combinatorics proof of identity for some higher-order recursive polynomial sequences and analyze the behavior of the minimal root of recursive polynomial sequences.

May.15-Jul.09 2022 MSU REU Program. Working as mentor assistant to help Prof Aklilu Zeleke to mentor undergraduate students Seijin Park, Etienne Phillips, Zhan Zhan, and Ilir Ziba. We worked on the Fibonacci like polynomials of multiple variables and found some identities and their relations with Bell polynomials, Fubini numbers, and Stirling numbers.

Spring 2022 MSU Exchange program. Helping Professor Preston Wake to mentor Yingdong Li. We worked on calculating the structure of Hecke algebra over complex fields and write a package for Sage.

Teaching

SummerA 2025 Teaching for MTH234(Cal3) section 103, grading, piazza

Fall 2024 Grading for MTH310(abstract algebra) and writing solutions for MTH327H(analysis)

Fall 2023 Teaching recitation for MTH234(Cal3) sections 005,022,031, grading, piazza and 3 MLC

Spring 2023 Leading TA. Organizing TA meeting for MTH235(Cal4), leading discussion, providing help, giving feedback.

Spring 2023 Teaching recitation for MTH299(Transition) sections 1,2. It is an introductory course for mathematical proof. Grading homework, writing solutions to homework, 1hr office hour per week, checking exam problems.

Fall 2022 Leading TA. Observing the recitations or labs taught by ULA and writing feedback of their teaching to math department

Fall 2022 Teaching recitation for MTH133(Cal2) sections 14,16,22, Grading and 4 MLC

Spring 2022 Grader for MTH411(Group theory)

Spring 2022 Teaching recitation for MTH133(Cal2, online),grading and 2 MLC.

Fall 2021 Teaching recitation for MTH235(Cal4,online) section0 07,008,009 and teaching special projects for honor section. Grading, MLC and Piazza

Spring 2021 Teaching recitation for MTH235(Cal4, online) for three sections.Grading,office hours and Piazza

Fall 2020 Teaching Assistant for qualifying course MTH818 Algebra1. Grading homework and one hour office hours per week

Services

Sep 2020-Aug 2021 **International student Liaison**, *AMS Graduate Student Chapter in MSU*
Coordinate the need of international student with the math department